ANSWER KEY & MARKING SCHEME SET 1

Chapter 12: **Beyond Earth** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 8 light-minutes	Approx.
2	**(d)** 60 kg	Mass does not change.
3	**(d)** Earth rotates on its axis	Tilt.
4	**(d)** waxing	Increasing.
5	**(b)** one sixth	Weaker gravity.
6	**(b)** stars appear to move across the night sky	Apparent motion.
7	**(d)** 30 March	About 29.5 days.
8	**(c)** 16	1440/90.
9	**(a)** Both A and R are true and R is the correct explanation of A.	A star makes its own light and heat, so R explains why the Sun is a star.
10	**(a)** Both A and R are true and R is the correct explanation of A.	The Moon only reflects the Sun's light.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	228/150 (1); about 1.5 cm (1).	1+1
12	They orbit the Sun (1); Closer than the stars (1).	1+1
13	Earth's rotation speed is highest (1); Gives extra boost (1).	1+1
14	24 hours = 1440 minutes (1); 1440/100 = 14.4 orbits (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	5000/10 (1); 500 days (1); Year/day (1).	1+1+1
16	Gravity pulls planets (1); Pull bends their path (1); Without it they would fly off in straight lines (1).	1+1+1
17	Moon is a quarter of the way around (1); Half of the lit side visible (1); Looks like a half disc (1).	1+1+1
	OR: 365/27 (1); About 13.5 (1); Approximately 13 complete orbits (1).	1+1+1
18	Long orbits (1); Take decades or centuries (1); Appear near the Sun (1).	1+1+1
	OR: Moon has no air or water (1); Little erosion (1); Earth's weather erases them (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) A constellation (1). (ii) The Pole star (1). (iii) Ursa Major (Saptarishi) (1). (iv) Starlight passes through the moving air of the atmosphere (1), which makes the light appear to flicker or twinkle (1).	1+1+1+2
20	New moon starts the cycle (1); First quarter day 8 (1); Full moon day 15 (1); Last quarter day 22 (1); Cycle about 29.5 days (1).	1+1+1+1+1
	OR: GPS (1); Medical imaging (1); Water purifiers (1); Materials (1); Weather prediction (1).	1+1+1+1+1

Mistakes to avoid:

- Writing that the Moon makes its own light; it only **reflects** sunlight.
- Writing nine planets or the wrong order; there are eight, starting with Mercury nearest to the Sun.
- Thinking the Moon itself changes shape; we only see different amounts of its lit half as it goes around the Earth.
- Mixing Chandrayaan (Moon) with Mangalyaan (Mars); Aryabhata was a satellite.
- Saying the Pole star is the brightest star or moves around; it appears nearly fixed and shows **north**.
- Confusing a star with a planet: the Sun is a star and the Earth is a planet.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

THE MASTERMIND COACHING CLASSES

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Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 2

Chapter 12: **Beyond Earth** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 90 minutes	42000/90.
2	**(d)** Mercury and Venus	Inner planets.
3	**(b)** west	Near the setting Sun.
4	**(d)** Mangalyaan	Mars Orbiter Mission.
5	**(a)** Jupiter	A storm.
6	**(d)** Venus	Evening star.
7	**(b)** Chandrayaan-3	India's success.
8	**(d)** Uranus	Tilt about 98 degrees.
9	**(a)** Both A and R are true and R is the correct explanation of A.	The moving air bends the light unevenly, causing twinkling.
10	**(d)** A is false but R is true.	A is false: there are eight planets. R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	It faces away from the Earth (1); It is not lit (1).	1+1
12	Cost and knowledge (1); Safety and cooperation (1).	1+1
13	380000/10000 (1); 38 cm (1).	1+1
14	Gravity is weaker (1); Mass stays the same (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	5 minutes = 300 s (1); 300 x 3 lakh (1); 9 crore km (1).	1+1+1
16	Far from the Sun (1); Less sunlight (1); Thick gases (1).	1+1+1
17	Collisions with satellites (1); Damage to space stations (1); Cost of replacement (1).	1+1+1
	OR: Sun is 150 million km away (1); Proxima is about 4 light years (1); Sun is nearer (1).	1+1+1
18	Scale the distances (1); Mark positions on a rope (1); Compare the spacing (1).	1+1+1
	OR: Communications (1); Remote sensing (1); Disaster warnings (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) New moon (1). (ii) Phases of the Moon (1). (iii) No, it reflects sunlight (1). (iv) Half of the Moon is always lit by the Sun (1); as the Moon moves around the Earth, we see different parts of the lit half (1).	1+1+1+2
20	Backup for humanity (1); Scientific research (1); No breathable air (1); Cold and radiation (1); Costly (1).	1+1+1+1+1
	OR: Used for direction (1); Calendar and seasons (1); Saptarishi (1); Observation records (1); Aryabhata used observations (1).	1+1+1+1+1

Mistakes to avoid:

- Writing that the Moon makes its own light; it only **reflects** sunlight.
- Writing nine planets or the wrong order; there are eight, starting with Mercury nearest to the Sun.
- Thinking the Moon itself changes shape; we only see different amounts of its lit half as it goes around the Earth.
- Mixing Chandrayaan (Moon) with Mangalyaan (Mars); Aryabhata was a satellite.
- Saying the Pole star is the brightest star or moves around; it appears nearly fixed and shows **north**.
- Confusing a star with a planet: the Sun is a star and the Earth is a planet.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 12: **Beyond Earth** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Jupiter, Earth, Mars, Mercury	Correct.
2	**(a)** rotate faster	Rotation period.
3	**(a)** atmosphere blurs the view	Air and light pollution.
4	**(b)** the Moon's history	Geology.
5	**(c)** Pole star lies almost along its extension	So the Pole star appears fixed.
6	**(c)** clear view without air disturbance	Better images.
7	**(d)** it can damage satellites	Collision risk.
8	**(c)** the Sun's bright light hides it	It remains.
9	**(c)** A is true but R is false.	A is true (launched in 1975); R is false, the Pole star is seen in the northern sky.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but the Indian missions do not explain the phases.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Thick carbon dioxide atmosphere (1); The atmosphere traps heat (1).	1+1
12	Earth rotates on its axis (1); Only the half facing the Sun is lit (1).	1+1
13	Atmospheric motion (1); Planets are discs (1).	1+1
14	It lies near the northern horizon (1); Earth's shape and latitude (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Eclipse: shadow of Earth or Moon (1); Phase: change in the lit part visible (1); Eclipse is occasional (1).	1+1+1
16	It moves fast sideways (1); Falls around the Earth (1); Orbit (1).	1+1+1
17	An optical illusion (1); Comparison with objects (1); Real size is the same (1).	1+1+1
	OR: Lower gravity (1); No air resistance (1); Less fuel needed (1).	1+1+1
18	They reflect sunlight (1); They are much closer than the stars (1); They shine steadily without twinkling much (1).	1+1+1
	OR: 75,000,000/25,000 (1); 3000 hours (1); About 125 days (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Aryabhata (1). (ii) Chandrayaan (1). (iii) A man-made object that goes around the Earth in space (1). (iv) Weather forecasting (1) and communication such as TV and phones (or navigation) (1).	1+1+1+2
20	Low orbits for imaging (1); High geostationary for TV (1); Medium for navigation (1); Different coverage (1); Different speeds (1).	1+1+1+1+1
	OR: Knowledge (1); Technology (1); Disaster management (1); Cost issues (1); Balance with needs (1).	1+1+1+1+1

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- Thinking the Moon itself changes shape; we only see different amounts of its lit half as it goes around the Earth.
- Mixing Chandrayaan (Moon) with Mangalyaan (Mars); Aryabhata was a satellite.
- Saying the Pole star is the brightest star or moves around; it appears nearly fixed and shows **north**.
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